TECHNICAL ASSESSMENT

AI/ML Developer Intern

Assessment type	Practical technical assignment
Recommended duration	2–3 days
Submission format	GitHub repository (source code, dataset, model, report, README)

Assessment Overview

Duration: 2–3 Days

Objective: Develop a prototype that uses machine learning to analyse cybersecurity assessment logs, detect suspicious activities, correlate security alerts, map findings to compliance requirements, and generate security reports using relevant data connectors.

Task 1: Dataset Preparation and Model Training

- Collect or create a cybersecurity log dataset containing normal and suspicious events.
- Clean, preprocess, label, and prepare the dataset for training.
- Train and evaluate an ML model for suspicious activity detection.
- Explain the model selection, performance metrics, false positives, and limitations.

Task 2: Security Log Analysis and Alert Correlation

- Analyse security assessment logs and identify suspicious patterns.
- Correlate multiple alerts using factors such as timestamps, source IPs, affected assets, event types, and severity.
- Group related alerts into potential security incidents.
- Explain the reasoning behind each detection using supporting evidence.

Task 3: Compliance Mapping

- Map detected findings to relevant controls from a framework such as ISO 27001 or NIST CSF.
- Include the associated control reference, supporting evidence, risk level, and recommended remediation for each finding.

Task 4: Data Connectors and Backend

- Integrate at least one relevant data source or connector to retrieve security logs, threat intelligence information, or assessment findings.
- Develop an API to process the collected data and return detection and correlation results.
- Implement input validation and basic error handling.

Task 5: Reporting and Presentation

- Generate a security report containing detected threats, correlated incidents, affected assets, severity, supporting evidence, compliance mappings, and remediation recommendations.
- Provide the source code, project structure, and setup instructions.
- Demonstrate the working prototype and explain the implemented workflow.

Submission Requirements

Candidates must submit the following:

- Source code and project structure.
- Dataset and details of the data sources used.
- Trained ML model and evaluation results.
- Working security log-analysis and alert-correlation pipeline.
- Compliance mapping and generated security report.
- README file containing setup instructions, execution steps, and known limitations.

GitHub Submission

After completing the assessment, upload the entire project to a GitHub repository and share the repository link for review. Ensure that the repository includes all necessary source files, documentation, and relevant project artefacts. The repository should be accessible to the reviewer. If it is private, grant the reviewer access. Do not upload passwords, API keys, access tokens, private keys, or other sensitive information to the repository.

Important Note

AI-generated explanations may be used to support the analysis, but they must not replace demonstrable detection logic, actual model evaluation, or verifiable security evidence. Candidates should be prepared to explain their implementation and demonstrate the working prototype.